

Q6 Three models are developed for predicting crop yield:

Model	Training MSE	Validation MSE
A	42	48
B	15	22
C	2	81

Analyse the bias-variance characteristics of the three models. Identify the likely underfitting and overfitting models and recommend the most appropriate model. Explain how increasing or decreasing model complexity, regularization, and additional training data could affect the observed results.

5

CO1

Q7 Consider the training examples shown in the Table below for a binary classification problem using ID3;

Instance No.	a1	a2	a3	Target Class
1	T	T	1	+
2	T	T	6	+
3	T	F	5	-
4	F	F	4	+
5	F	T	7	-
6	F	T	3	-
7	F	F	8	-
8	T	F	7	+
9	F	T	5	-

1. What is the entropy of this collection of training examples with respect to the positive class?
2. What are the information gains of a1 and a2 relative to these training examples?
3. For a3, which is a continuous attribute, compute the information gain for every possible split.
4. What is the best split (between a1 and a2) according to the information gain?

(1+2+1+1)
5

CO3

SECTION-C

Q8 Consider the following dataset representing study hours x and examination marks y:

Study Hours x	Marks y
1	2
2	3
3	5
4	6

Assume the regression model:

$$\hat{y} = w_0 + w_1x$$

- a) Compute the regression equation from the provided dataset.
- b) Write the Mean Squared Error cost function for the given model.
- c) Using the obtained regression equation, predict y when x = 5.
- d) Explain how repeated gradient-descent iterations improve the regression model.

(2+1+2+2)
7

CO2

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Odd Semester, Mid Term Examination, Academic Session: 2026-2027
FoSTA, School of Computer Science and Engineering
Department of Artificial Intelligence and Machine Learning (AIML)
BTech -CSE(AIML)

Course Code: AIM3101

Course: Machine Learning

Semester: V

Time: 1.5 hrs.

Max. Marks: 30

Instructions: All questions are compulsory.

Missing data, if any, may be assumed suitably.

Calculator is allowed.

SECTION A

S.No.		Marks	CO										
Q 1	Using the Manhattan and Euclidean distance measures, determine the nearest point to Y (7,5) among the following points: $X_1 = (5,1)$; $X_2 = (6,7)$; and $X_3 = (4,5)$	3	CO1										
Q 2	A logistic regression classifier uses; $P(Y = 1 X) = \frac{1}{1 + e^{-(2x_1 - 3x_2 + 1)}}$ A sample has $x_1=2$ and $x_2=1$. If the classification threshold is changed from 0.5 to 0.9, which outcome is most likely? A. The predicted probability changes B. The model parameters change C. The predicted class may change from 1 to 0, although probability remains unchanged D. The sigmoid function changes	2	CO2										
Q 3	Illustrate the main difference between Precision and Recall?	2	CO3										
Q 4	Match the situation with the most appropriate technique:	2	CO2										
	<table border="0"> <thead> <tr> <th>Situation</th> <th>Technique</th> </tr> </thead> <tbody> <tr> <td>A. Many irrelevant variables should be eliminated.</td> <td>1. KNN.</td> </tr> <tr> <td>B. Strong multicollinearity but all predictors should remain.</td> <td>2. Lasso.</td> </tr> <tr> <td>C. Nonlinear decision boundary required.</td> <td>3. Ridge.</td> </tr> <tr> <td>D. Instance-based prediction required.</td> <td>4. Kernel SVM</td> </tr> </tbody> </table>	Situation	Technique	A. Many irrelevant variables should be eliminated.	1. KNN.	B. Strong multicollinearity but all predictors should remain.	2. Lasso.	C. Nonlinear decision boundary required.	3. Ridge.	D. Instance-based prediction required.	4. Kernel SVM		
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SECTION B

Q 5	Consider the data set shown in the following Table:	(2+2) 4	CO3																																																							
<table border="1"><thead><tr><th>Record</th><th>A</th><th>B</th><th>C</th><th>Class</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>+</td></tr><tr><td>2</td><td>0</td><td>0</td><td>1</td><td>-</td></tr><tr><td>3</td><td>0</td><td>1</td><td>1</td><td>-</td></tr><tr><td>4</td><td>0</td><td>1</td><td>1</td><td>-</td></tr><tr><td>5</td><td>0</td><td>0</td><td>1</td><td>+</td></tr><tr><td>6</td><td>1</td><td>0</td><td>1</td><td>+</td></tr><tr><td>7</td><td>1</td><td>0</td><td>1</td><td>-</td></tr><tr><td>8</td><td>1</td><td>0</td><td>1</td><td>-</td></tr><tr><td>9</td><td>1</td><td>1</td><td>1</td><td>+</td></tr><tr><td>10</td><td>1</td><td>0</td><td>1</td><td>+</td></tr></tbody></table>		Record	A	B	C	Class	1	0	0	0	+	2	0	0	1	-	3	0	1	1	-	4	0	1	1	-	5	0	0	1	+	6	1	0	1	+	7	1	0	1	-	8	1	0	1	-	9	1	1	1	+	10	1	0	1	+		
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a. Estimate the conditional probabilities for $P(A +)$, $P(B +)$, $P(C +)$, $P(A -)$, $P(B -)$, and $P(C -)$. b. Use the estimate of conditional probabilities given in the question to predict the class label for a test sample ($A = 0$, $B = 1$, $C = 0$) using the Naive Bayes approach.																																																										